

for
statements

CSCI
134



a (possibly apocryphal)
story about german
mathematician **carl**
gauss as a child

what is the sum
of the numbers
from 1 to 100?



what is the sum
of the numbers
from 1 to 100?

5050



what is the sum
of the numbers
from 1 to 100?

5050



$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$$



$$\begin{array}{r} 1 + 2 + 3 + 4 + 5 \\ 10 + 9 + 8 + 7 + 6 \end{array} +$$



$$\begin{array}{r}
 1 + 2 + 3 + 4 + 5 \\
 10 + 9 + 8 + 7 + 6 \\
 \hline
 11 \quad 11 \quad 11 \quad 11 \quad 11
 \end{array}$$





$$\begin{array}{r} 1+2+3+4+5 \\ 10+9+8+7+6 \\ \hline 11 \quad 11 \quad 11 \quad 11 \quad 11 \\ \hline \end{array} = 55$$

5×11

$$1 + 2 + 3 + \dots + 98 + 99 + 100$$



$$1 + 2 + 3 + \dots + 50 +$$
$$100 + 99 + 98 + \dots + 51$$



$$\begin{array}{r}
 1 + 2 + 3 + \dots + 50 \\
 100 + 99 + 98 + \dots + 51 \\
 \hline
 101 \quad 101 \quad 101 \quad \quad 101
 \end{array}$$

$$\underbrace{\hspace{15em}}_{50 \times 101} = 5050$$



ok, but what if we aren't a
german math prodigy?

```
In [1]: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10
```

```
Out[1]: 55
```

```
In [2]: 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 +  
...: 14 + 15 + 16 + 17 + 18 + 19 + 20 + 21 + 22 + 23 + 24 +  
...: 25 + 26 + 27 + 28 + 29 + 30 + 31 + 32 + 33 + 34 + 35 +  
...: 36 + 37 + 38 + 39 + 40 + 41 + 42 + 43 + 44 + 45 + 46 +  
...: 47 + 48 + 49 + 50 + 51 + 52 + 53 + 54 + 55 + 56 + 57 +  
...: 58 + 59 + 60 + 61 + 62 + 63 + 64 + 65 + 66 + 67 + 68 +  
...: 69 + 70 + 71 + 72 + 73 + 74 + 75 + 76 + 77 + 78 + 79 +  
...: 80 + 81 + 82 + 83 + 84 + 85 + 86 + 87 + 88 + 89 + 90 +  
...: 91 + 92 + 93 + 94 + 95 + 96 + 97 + 98 + 99 + 100
```

```
Out[2]: 5050
```

this is a lot of typing

```
In [1]: total = 0
```

```
In [2]: for number in range(1, 101):  
...:     total = total + number  
...:
```

```
In [3]: print(total)  
5050
```

but we can also do this

for every integer in **this range**



```
for variable in range(int, int):
```

python code



this code is executed

For every integer from **this** up to (but not including) **this**

```
for variable in range(int, int):
```

python code

this code is executed

this code

```
for num in range(1, 5):  
    print(num)
```

is equivalent
to this code

```
num = 1  
print(num)  
num = 2  
print(num)  
num = 3  
print(num)  
num = 4  
print(num)
```

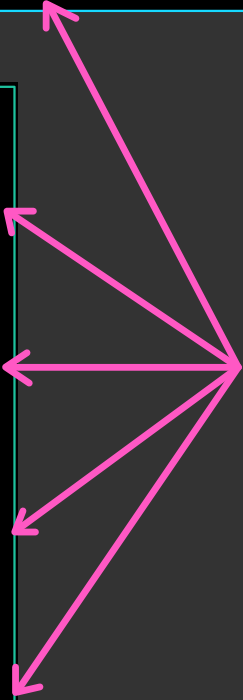
this code

```
for num in range(1, 5):  
    print(num)
```

is equivalent
to this code

```
num = 1  
print(num)  
num = 2  
print(num)  
num = 3  
print(num)  
num = 4  
print(num)
```

"iterations"



this code

```
for num in range(1, 5):  
    print(num)
```

is equivalent
to this code

```
num = 1  
print(num)  
num = 2  
print(num)  
num = 3  
print(num)  
num = 4  
print(num)
```

up to but
not including



this code

```
total = 0
for num in range(1, 5):
    total = total + num
```

is equivalent
to this code

```
total = 0
num = 1
total = total + num
num = 2
total = total + num
num = 3
total = total + num
num = 4
total = total + num
```

accumulation
variable

(builds up a
value as the
iterations
proceed)

```
total = 0
```

```
for num in range(1, 5):  
    total = total + num
```

```
total = 0
```

```
num = 1
```

```
total = total + num
```

```
num = 2
```

```
total = total + num
```

```
num = 3
```

```
total = total + num
```

```
num = 4
```

```
total = total + num
```

variants

from 0 up to
(but not including) 5

```
In [1]: for num in range(5):  
...:     print(num)  
...:  
0  
1  
2  
3  
4
```

from 1 up to
(but not including) 9
counting by 2

```
In [1]: for num in range(1, 9, 2):  
...:     print(num)  
...:  
1  
3  
5  
7
```

from 5 down to
(but not including) 1
counting by -1

```
In [1]: for num in range(5, 1, -1):  
...:     print(num)  
...:  
5  
4  
3  
2
```

variants

from 0 up to
(but not including) 5

```
In [1]: for num in range(5):  
...:     print(num)  
...:  
0  
1  
2  
3  
4
```

from 1 up to
(but not including) 9
counting by 2

```
In [1]: for num in range(1, 9, 2):  
...:     print(num)  
...:  
1  
3  
5  
7
```

"stride"

from 5 down to
(but not including) 1
counting by -1

```
In [1]: for num in range(5, 1, -1):  
...:     print(num)  
...:  
5  
4  
3  
2
```


statements so far

assignment

variable = expression

def

def name (arguments):

4 spaces

return

return value

(what the Function computes)

for

for variable in range(int, int):

python code